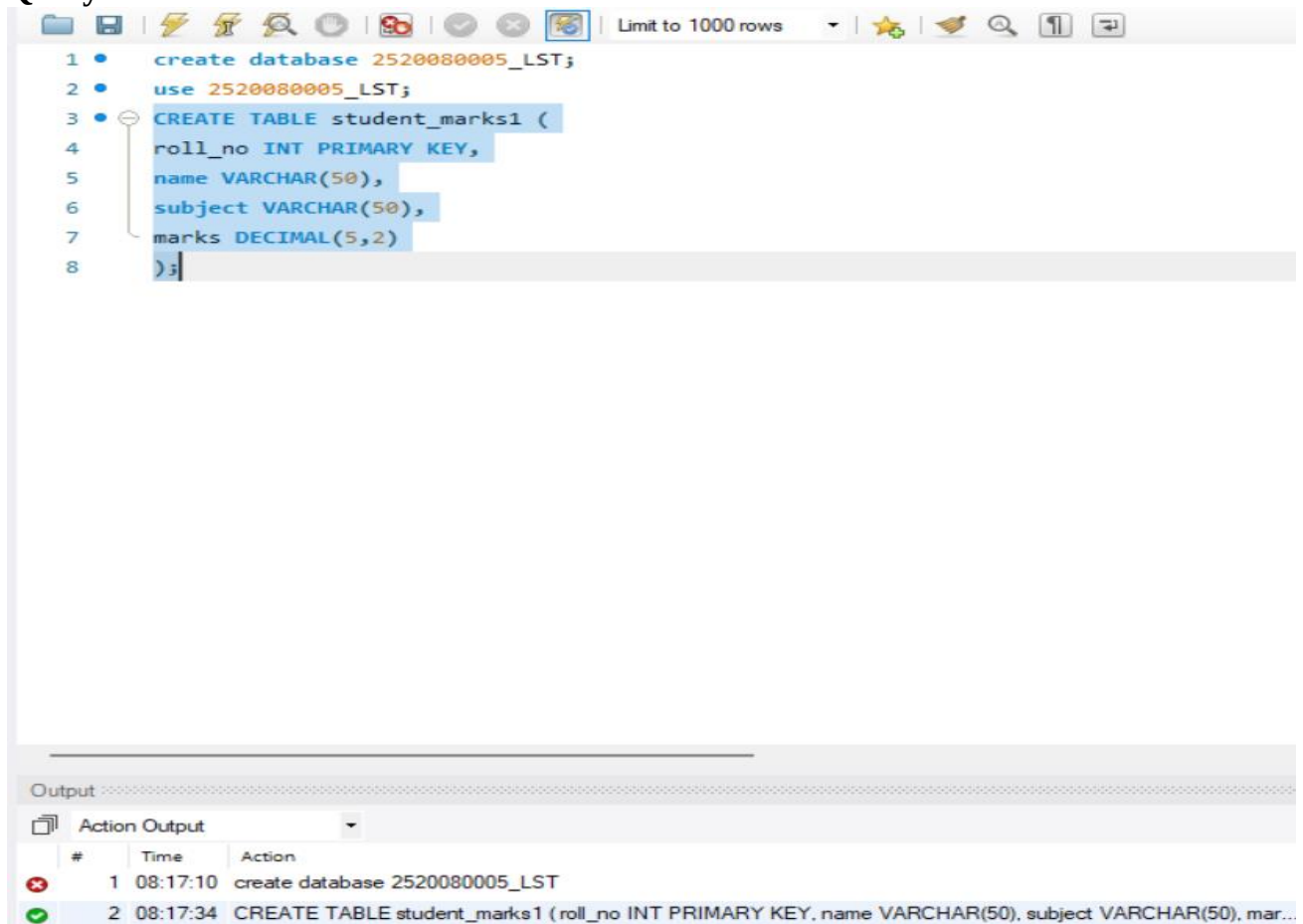


SKILL WEEK-1

Query 1 : Create Table



The screenshot shows a database query editor interface. The top toolbar includes icons for file operations, execution, and search. The main text area contains the following SQL code:

```
1 • create database 2520080005_LST;  
2 • use 2520080005_LST;  
3 • CREATE TABLE student_marks1 (  
4   roll_no INT PRIMARY KEY,  
5   name VARCHAR(50),  
6   subject VARCHAR(50),  
7   marks DECIMAL(5,2)  
8 );
```

Below the query editor is an "Output" section with a dropdown menu set to "Action Output". It displays a log of executed actions:

#	Time	Action
1	08:17:10	create database 2520080005_LST
2	08:17:34	CREATE TABLE student_marks1 (roll_no INT PRIMARY KEY, name VARCHAR(50), subject VARCHAR(50), mar...

Query 2 : Insert Data

```
1 • create database 2520080005_LST;
2 • use 2520080005_LST;
3 • CREATE TABLE student_marks1 (
4   roll_no INT PRIMARY KEY,
5   name VARCHAR(50),
6   subject VARCHAR(50),
7   marks DECIMAL(5,2)
8 );
9
10 • INSERT INTO student_marks1 (roll_no, name, subject, marks) VALUES
11   (1, 'Ravi', 'Math', 85.50),
12   (2, 'Sita', 'Math', 92.75),
13   (3, 'Anil', 'Math', 78.40),
14   (4, 'Priya', 'Math', 88.90),
15   (5, 'Vijay', 'Math', 80.25),
16   (6, 'subbusir', 'aws', 98.50),
17   (7, 'dwaraka', 'dbms', 95.75),
18   (8, 'ranjani', 'english', 97.40),
19   (9, 'kavithamam', 'aws1', 99.90),
20   (10, 'seetha', 'azure', 82.25);
21
```

Output			
Action Output			
#	Time	Action	
✖ 1	08:17:10	create database 2520080005_LST	
✔ 2	08:17:34	CREATE TABLE student_marks1 (roll_no INT PRIMARY KEY, name VARCHAR(50), subject VARCHAR(50), mar...	
✖ 3	08:20:15	INSERT INTO student_marks (roll_no, name, subject, marks) VALUES (1, 'Ravi', 'Math', 85.50), (2, 'Sita', 'Math', 9...	
✔ 4	08:20:26	INSERT INTO student_marks1 (roll_no, name, subject, marks) VALUES (1, 'Ravi', 'Math', 85.50), (2, 'Sita', 'Math', ...	

Query 3 : Simple select

```
16      (6, 'subbusir', 'aws', 98.50),
17      (7, 'dwaraka', 'dbms', 95.75),
18      (8, 'ranjani', 'english', 97.40),
19      (9, 'kavithamam', 'aws1', 99.90),
20      (10, 'seetha', 'azure', 82.25);
21
22 • SELECT * FROM student_marks1;
23
24
```

Result Grid				
Filter Rows:				
Edit:   				
Export/Import:  				
Wrap Cell Content: 				
	roll_no	name	subject	marks
▶	1	Ravi	Math	85.50
	2	Sita	Math	92.75
	3	Anil	Math	78.40
	4	Priya	Math	88.90
	5	Vijay	Math	80.25
	6	subbusir	aws	98.50
	7	dwaraka	dbms	95.75
	8	ranjani	english	97.40
	9	kavithamam	aws1	99.90
	10	seetha	azure	82.25
•	NULL	NULL	NULL	NULL

dent_marks1 1 x

Output



Action Output

	#	Time	Action
✓	2	08:17:34	CREATE TABLE student_marks1 (roll_no INT PRIMARY KEY, name VARCHAR(50), subject VARCHAR(50), ...
✗	3	08:20:15	INSERT INTO student_marks (roll_no, name, subject, marks) VALUES (1, 'Ravi', 'Math', 85.50), (2, 'Sita', 'Math...
✓	4	08:20:26	INSERT INTO student_marks1 (roll_no, name, subject, marks) VALUES (1, 'Ravi', 'Math', 85.50), (2, 'Sita', 'Mat...
✗	5	08:21:24	SELECT * FROM student_marks LIMIT 0, 1000
✓	6	08:21:32	SELECT * FROM student_marks1 LIMIT 0, 1000

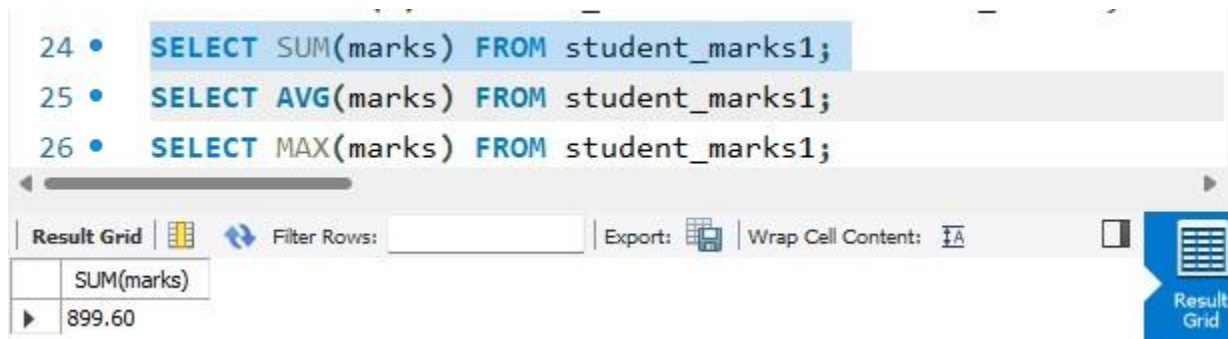
Query 4 : Aggregate Functions

1. `SELECT COUNT(*) AS total_students FROM student_marks1;`

The screenshot displays a database management interface. At the top, a SQL query is entered in a text area: `SELECT COUNT(*) AS total_students FROM student_marks1;`. Below the query area, a toolbar includes options for 'Result Grid', 'Filter Rows', 'Exports', and 'Wrap Cell Content'. The 'Result Grid' is currently selected, showing a single row with the column name 'total_students' and the value '10'. To the right of the main workspace, a vertical sidebar contains icons for 'Result Grid', 'Form Editor', and 'Field Types'. At the bottom of the interface, an 'Output' pane is visible, showing a log of database actions. The log includes several 'INSERT INTO' and 'SELECT *' statements, some of which resulted in error messages (Error Code: 1146: Table 'sys.student_marks' doesn't exist). The final action in the log is the execution of the query shown above, which returned 1 row(s).

#	Time	Action	Message	Duration / Fetch
3	08:20:15	INSERT INTO student_marks (roll_no, name, subject, marks) VALUES (1, 'Ravi', 'Math', 85.50), (2, 'Sita', 'Math...	Error Code: 1146: Table 'sys.student_marks' doesn't exist	0.047 sec
4	08:20:26	INSERT INTO student_marks1 (roll_no, name, subject, marks) VALUES (1, 'Ravi', 'Math', 85.50), (2, 'Sita', 'Mat...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.063 sec
5	08:21:24	SELECT * FROM student_marks LIMIT 0, 1000	Error Code: 1146: Table 'sys.student_marks' doesn't exist	0.000 sec
6	08:21:32	SELECT * FROM student_marks1 LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
7	09:34:26	SELECT COUNT(*) AS total_students FROM student_marks1 LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.000 sec

2. SELECT SUM(marks) FROM student_marks1;



24 • SELECT SUM(marks) FROM student_marks1;

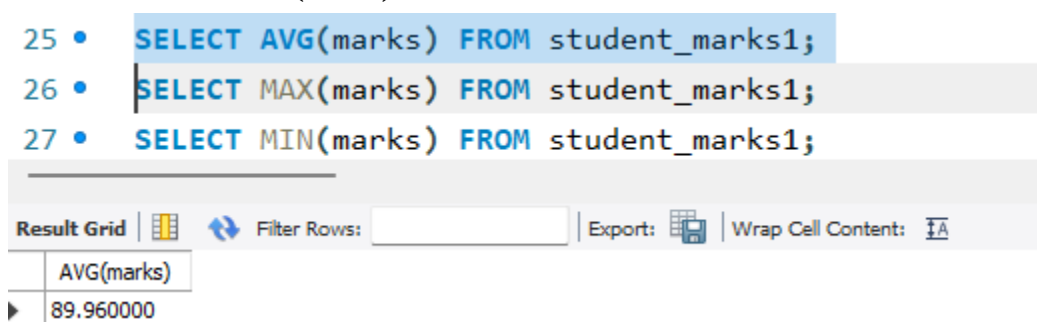
25 • SELECT AVG(marks) FROM student_marks1;

26 • SELECT MAX(marks) FROM student_marks1;

Result Grid

SUM(marks)
899.60

3. SELECT AVG(marks) FROM student_marks1;



25 • SELECT AVG(marks) FROM student_marks1;

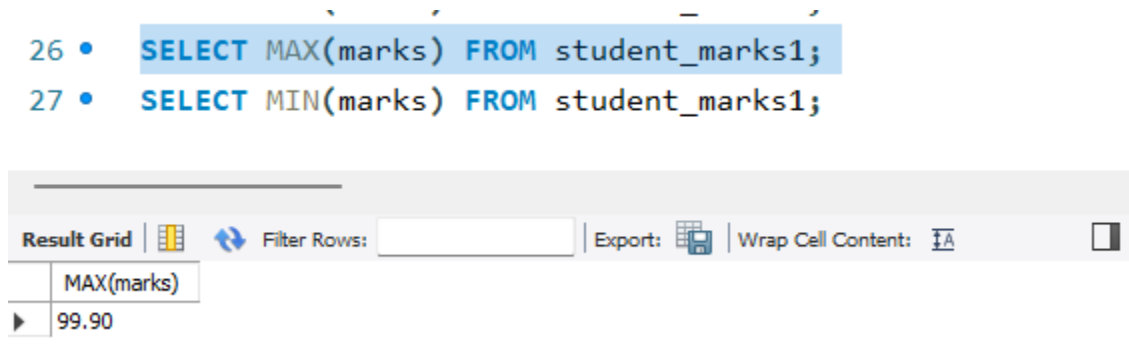
26 • SELECT MAX(marks) FROM student_marks1;

27 • SELECT MIN(marks) FROM student_marks1;

Result Grid

AVG(marks)
89.960000

4. SELECT MAX(marks) FROM student_marks1;



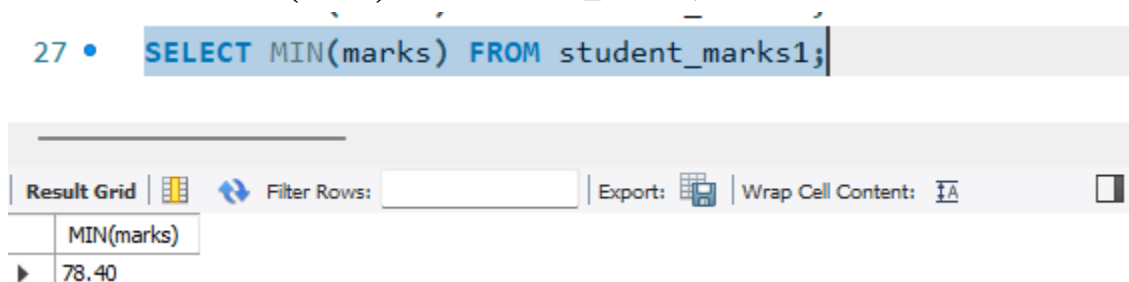
26 • SELECT MAX(marks) FROM student_marks1;

27 • SELECT MIN(marks) FROM student_marks1;

Result Grid

MAX(marks)
99.90

5. SELECT MIN(marks) FROM student_marks1;



27 • SELECT MIN(marks) FROM student_marks1;

Result Grid

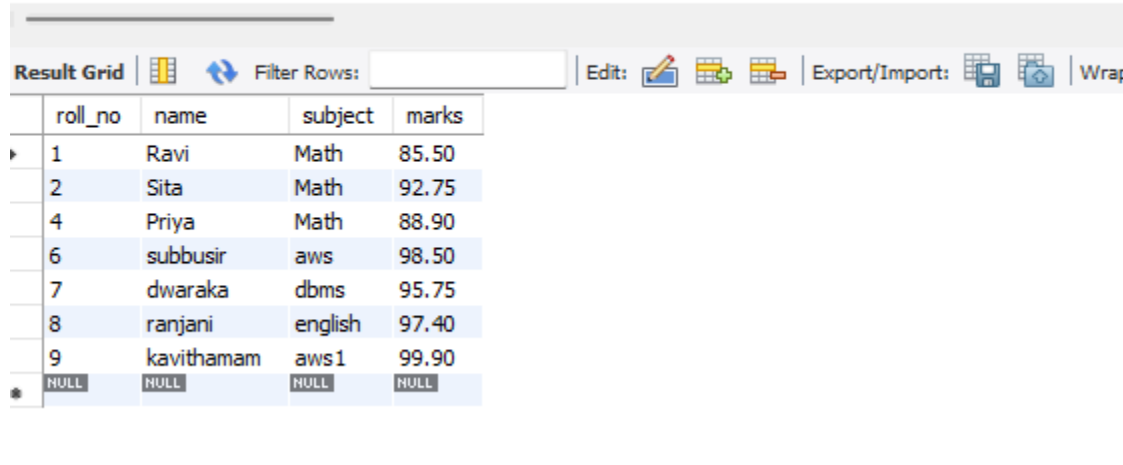
MIN(marks)
78.40

Query 5 : Filtering with WHERE

1. SELECT * FROM student_marks WHERE marks > 85;

```
29 • SELECT * FROM student_marks1 WHERE marks > 85;
```

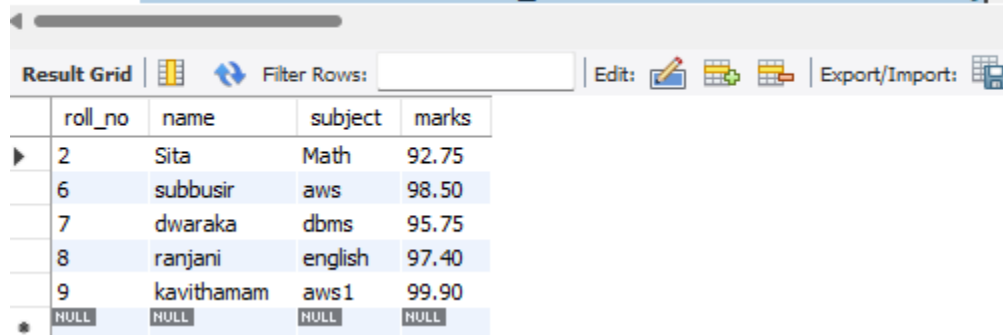
```
30 • SELECT * FROM student_marks1 WHERE marks >= 90;
```



roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
4	Priya	Math	88.90
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90
NULL	NULL	NULL	NULL

2. SELECT * FROM student_marks1 WHERE marks >= 90;

```
30 • SELECT * FROM student_marks1 WHERE marks >= 90;
```

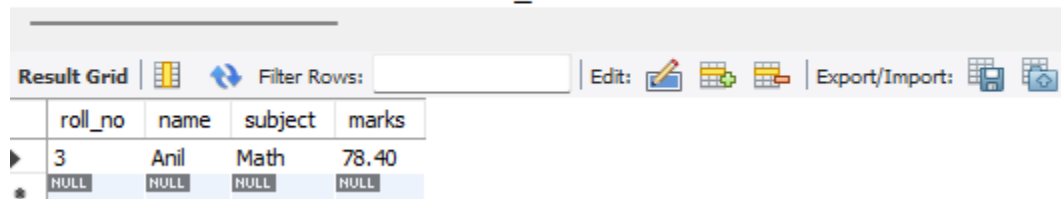


roll_no	name	subject	marks
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90
NULL	NULL	NULL	NULL

3. SELECT * FROM student_marks1 WHERE marks < 80;

```
31 • SELECT * FROM student_marks1 WHERE marks < 80;
```

```
32 • SELECT * FROM student_marks1 WHERE marks BETWEEN 80
```



roll_no	name	subject	marks
3	Anil	Math	78.40
NULL	NULL	NULL	NULL

4. SELECT * FROM student_marks1 WHERE marks BETWEEN 80 AND 90;

32 • **SELECT * FROM student_marks1 WHERE marks BETWEEN 80 AND 90;**

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

	roll_no	name	subject	marks
▶	1	Ravi	Math	85.50
	4	Priya	Math	88.90
	5	Vijay	Math	80.25
	10	seetha	azure	82.25
*	NULL	NULL	NULL	NULL

Result Grid
Form Editor

5. SELECT * FROM student_marks1 WHERE name LIKE 'P%';

33 • **SELECT * FROM student_marks1 WHERE name LIKE 'P%';**

34 • **SELECT * FROM student_marks1 WHERE name IN ('Ravi', 'Sita', 'Vijay');**

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

	roll_no	name	subject	marks
▶	4	Priya	Math	88.90
*	NULL	NULL	NULL	NULL

6. SELECT * FROM student_marks1 WHERE name IN ('Ravi', 'Sita', 'Vijay');

34 • **SELECT * FROM student_marks1 WHERE name IN ('Ravi', 'Sita', 'Vijay');**

35 • **SELECT * FROM student_marks1 WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'P%');**

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	roll_no	name	subject	marks
▶	1	Ravi	Math	85.50
	2	Sita	Math	92.75
	5	Vijay	Math	80.25
*	NULL	NULL	NULL	NULL

7. SELECT * FROM student_marks1 WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'P%');

35 • **SELECT * FROM student_marks1 WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'P%');**

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	roll_no	name	subject	marks
▶	1	Ravi	Math	85.50
	2	Sita	Math	92.75
	4	Priya	Math	88.90
*	NULL	NULL	NULL	NULL

Query 6 : Update Queries

1. UPDATE student_marks SET marks = 90.00 WHERE roll_no = 3;

```
36 • UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3;
37 • UPDATE student_marks1 SET subject = 'Remedial Math' WHERE marks < 80;
38
39
40
```

Output		
Action Output		
#	Time	Message
17	10:57:21	CREATE TABLE student_marks1 (roll_no INT PRIMARY KEY, name VARCHAR(50), subject VARCHAR(50), marks DECIMAL(5,2))
18	10:57:36	SELECT * FROM student_marks1 WHERE name LIKE 'P%' LIMIT 0, 1000
19	10:58:07	SELECT * FROM student_marks1 WHERE name IN ('Ravi', 'Sita', 'Vijay') LIMIT 0, 1000
20	10:58:39	SELECT * FROM student_marks1 WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'P%') LIMIT 0, 1000
21	11:09:59	UPDATE student_marks SET marks = 90.00 WHERE roll_no = 3
22	11:10:11	UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3

2. UPDATE student_marks1 SET subject = 'Remedial Math' WHERE marks < 80;

```
37 • UPDATE student_marks1 SET subject = 'Remedial Math' WHERE marks < 80;
38
39
40
```

Output		
Action Output		
#	Time	Message
22	11:10:11	UPDATE student_marks1 SET marks = 90.00 WHERE roll_n... 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
23	11:10:45	UPDATE student_marks1 SET subject = 'Remedial Math' WH... Error Code: 1175. You are using safe update mode and you tri.
24	11:43:50	SET SQL_SAFE_UPDATES = 0 0 row(s) affected
25	11:44:00	UPDATE student_marks1 SET marks = 90.00 WHERE roll_n... 0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0
26	11:44:15	SELECT * FROM student_marks1 LIMIT 0, 1000 10 row(s) returned
27	11:45:16	UPDATE student_marks1 SET subject = 'Remedial Math' WH... 0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0

Query 7 : Delete Queries

1. DELETE FROM student_marks WHERE roll_no = 5;

38

39 • DELETE FROM student_marks1 WHERE roll_no=5;

40 • DELETE FROM student_marks1 WHERE marks<75;

41

42

Output				
Action Output				
#	Time	Action	Message	
✓ 24	11:43:50	SET SQL_SAFE_UPDATES = 0	0 row(s) affected	
✓ 25	11:44:00	UPDATE student_marks1 SET marks = 90.00 WHERE roll_n...	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	
✓ 26	11:44:15	SELECT * FROM student_marks1 LIMIT 0, 1000	10 row(s) returned	
✓ 27	11:45:16	UPDATE student_marks1 SET subject = 'Remedial Math' WH...	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0	
✗ 28	11:47:09	DELETE FROM student_marks WHERE roll_no = 5	Error Code: 1146. Table '2520080069_ananya.student_marks...	
✓ 29	11:48:50	DELETE FROM student_marks1 WHERE roll_no=5	1 row(s) affected	

2. DELETE FROM student_marks1 WHERE marks<75;

40 • DELETE FROM student_marks1 WHERE marks<75;

41

42

Output				
Action Output				
#	Time	Action	Message	
✓ 25	11:44:00	UPDATE student_marks1 SET marks = 90.00 WHERE roll_n...	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	
✓ 26	11:44:15	SELECT * FROM student_marks1 LIMIT 0, 1000	10 row(s) returned	
✓ 27	11:45:16	UPDATE student_marks1 SET subject = 'Remedial Math' WH...	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0	
✗ 28	11:47:09	DELETE FROM student_marks WHERE roll_no = 5	Error Code: 1146. Table '2520080069_ananya.student_marks...	
✓ 29	11:48:50	DELETE FROM student_marks1 WHERE roll_no=5	1 row(s) affected	
✓ 30	11:49:07	DELETE FROM student_marks1 WHERE marks<75	0 row(s) affected	

Query 8 : Sorting

1. SELECT * FROM student_marks ORDER BY marks ASC;

```
42 • SELECT * FROM student_marks1 ORDER BY marks ASC;
43 • SELECT * FROM student_marks1 ORDER BY marks DESC;
44 • SELECT * FROM student_marks1 ORDER BY name ASC;
```

Result Grid

 Filter Rows:

Edit: 

Export/Import:  

	roll_no	name	subject	marks
	10	seetha	azure	82.25
	1	Ravi	Math	85.50
	4	Priya	Math	88.90
	3	Anil	Math	90.00
	2	Sita	Math	92.75
	7	dwaraka	dbms	95.75
	8	ranjani	english	97.40
	6	subbusir	aws	98.50
	9	kavithamam	aws1	99.90
	NULL	NULL	NULL	NULL

2. SELECT * FROM student_marks1 ORDER BY marks DESC;

```
43 • SELECT * FROM student_marks1 ORDER BY marks DESC;
44 • SELECT * FROM student_marks1 ORDER BY name ASC;
45 • SELECT * FROM student_marks1 ORDER BY name DESC;
```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap

	roll_no	name	subject	marks
▶	9	kavithamam	aws1	99.90
	6	subbusir	aws	98.50
	8	ranjani	english	97.40
	7	dwaraka	dbms	95.75
	2	Sita	Math	92.75
	3	Anil	Math	90.00
	4	Priya	Math	88.90
	1	Ravi	Math	85.50
	10	seetha	azure	82.25
✱	NULL	NULL	NULL	NULL

3. `SELECT * FROM student_marks1 ORDER BY name ASC;`

```
44 • SELECT * FROM student_marks1 ORDER BY name ASC;
```

```
45 • SELECT * FROM student_marks1 ORDER BY name DESC;
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wr
roll_no	name	subject	marks	
3	Anil	Math	90.00	
7	dwaraka	dbms	95.75	
9	kavithamam	aws1	99.90	
4	Priya	Math	88.90	
8	ranjani	english	97.40	
1	Ravi	Math	85.50	
10	seetha	azure	82.25	
2	Sita	Math	92.75	
6	subbusir	aws	98.50	
NULL	NULL	NULL	NULL	

4. `SELECT * FROM student_marks1 ORDER BY name DESC;`

```
45 • SELECT * FROM student_marks1 ORDER BY name DESC;
```

46

Result Grid				
			Filter Rows:	
			Edit:	
roll_no	name	subject	marks	
6	subbusir	aws	98.50	
2	Sita	Math	92.75	
10	seetha	azure	82.25	
1	Ravi	Math	85.50	
8	ranjani	english	97.40	
4	Priya	Math	88.90	
9	kavithamam	aws1	99.90	
7	dwaraka	dbms	95.75	
3	Anil	Math	90.00	
* NULL	NULL	NULL	NULL	

Query 9 : GROUP BY

1. SELECT subject, SUM(marks) AS total_marks FROM student_marks GROUP BY subject;

```
48 • SELECT subject, SUM(marks) AS total_marks FROM student_marks1 GROUP BY subject;
49 • SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
subject	total_marks		
Math	357.15		
aws	98.50		
dbms	95.75		
english	97.40		
aws1	99.90		
azure	82.25		

2. SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject;

```
49 • SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject;
50 • SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY subject;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
subject	avg_marks		
Math	89.287500		
aws	98.500000		
dbms	95.750000		
english	97.400000		
aws1	99.900000		
azure	82.250000		

3. SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY subject;

```
50 • SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY subject;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
subject	num_students		
Math	4		
aws	1		
dbms	1		
english	1		
aws1	1		
azure	1		

Query 10 : Having

1. SELECT subject, AVG(marks) AS avg_marks FROM student_marks GROUP BY subject HAVING AVG(marks) > 90;

```
52 • SELECT subject, AVG(marks) AS avg_marks
53 FROM student_marks1
54 GROUP BY subject
55 HAVING AVG(marks) > 90;
56
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
	subject	avg_marks	
▶	aws	98.500000	
	dbms	95.750000	
	english	97.400000	
	aws1	99.900000	

2. SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY subject HAVING COUNT(*) > 1;

```
57 • SELECT subject, COUNT(*) AS num_students
58 FROM student_marks1
59 GROUP BY subject
60 HAVING COUNT(*) > 1;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
	subject	num_students	
	Math	4	